SIMATS ENGINEERING

ASSESSMENT TOOL 1: Short Answer Test

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions.* (BL2)

Assessment Tool Weightage: 40%

Dave's Taxonomy: Imitation (Level 1) **Total Marks: 25**

Question Count: 5

SHORT ANSWER TEST

Code of Conduct

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: _____

Assessment Tasks

Short Answer Test

1. Define cloud computing and explain any four characteristics that make it different from traditional on-premise computing.
2. Trace the evolution of cloud computing from hardware evolution to internet software evolution, and show how these changes enabled modern cloud platforms.
3. Explain the role of server virtualization in cloud computing. Differentiate between virtualization as an enabling technology and cloud computing as a service model.
4. Discuss the role of **data centers** in delivering cloud services..
5. Differentiate IaaS, PaaS, SaaS, and XaaS with one suitable real-world example for each model.

Short Answer Test Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Conceptual Accuracy	30%	Accurate definitions and precise explanation of all key terms	Mostly accurate with minor gaps	Partial understanding with noticeable errors	Incorrect or irrelevant explanation
Coverage of Concepts	20%	Covers all expected sub-points completely	Covers most expected points	Covers only basic points	Very limited coverage
Use of Examples	15%	Uses highly relevant cloud examples to strengthen the answer	Uses relevant examples with minor mismatch	Uses vague or partially relevant examples	No useful examples
Comparison / Differentiation	20%	Clearly distinguishes concepts with strong logic	Reasonable differentiation with minor overlap	Comparison is weak or incomplete	Fails to differentiate concepts
Clarity of Expression	15%	Clear, organized, and concise answer writing	Generally clear with few issues	Understandable but poorly organized	Difficult to follow

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Course: Cloud computing
Short-Answer test

① Define cloud computing and Explain four characteristics:

:) Definition:

cloud computing is the delivery of computing services such as server storage, database, networking and software through the Internet instead of using local computer

Four characteristics:

i) on-demand self service where user can get computing resource when required without Human
ii) Cloud services can be accessed through the Internet using mobile, laptops and other device
iii) cloud provider shared computing resource among many user efficiently

② Evolution of cloud computing:

cloud computing developed through several stages.

:) Hard evolution:

computer become smaller, faster and cheaper. This improved computing capacity

:) Mainframe computing:

Many user accessed a single powerful mainframe computer through terminals

:) Client - Server computing:

Processing was divided between client computers and servers connected through network

:) Virtualization:

one physical server could run multiple virtual machine (VMs) using a software layer called as hypervisor.

(3) Role of Server virtualization:

Server virtualization divide one physical server into multiple virtual machine (VMs).

Role in cloud computing:

- Improve server utilization
- Reduce hardware cost
- Allow multiple user to share resource
- Provides flexibility and scalability
- makes resource allocation easier

Virtualization vs cloud computing:

• Technology	• Server delivery model
• creates virtual resource	• Delivers computing services
• Can work without Internet	• Usually accessed through network

Virtualization is an enabling technology, while cloud computing uses such technologies to provide services of user.

④ Role of data centres in cloud services:

A data centre is a facility containing servers, storage system, networking equipment and other infrastructure

Roles:

- Computing - Provides servers to run application
- Storage - store user data and application
- Networking - connect cloud resource to user through the Internet
- Security - Protects data using firewalls, authentication and monitoring
- Backup - Maintain copies of data to prevent data lost

Data centre are the physical foundation of cloud computing they provide the hardwork and infrastructure required to delivers cloud services.

Iaas, Paas, Saas and Xaas

cloud computing provides different services based on the requirements

① Iaas → infrastructure as a service:

Provide Virtual service, storage and networking through the Internet

Ex - Amazon EC2

② Paas → Platform as a service:

Provide Virtual server storage and networking through the Internet test deploy applications.

Ex - Google App Engine

③ Saas → Software as a service:

Provide ready-to-use software through the internet user not install or maintain it

④ Xaas → Anything as a service:

Provide different it services through the cloud, such as database, storage and security services. Ex → Amazon RDS.